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Part I

preliminaries

1. Basics of Magnetism

1.1 Direct Exchange

1.1.1 Coulomb exchange

We are looking for a mechanism that can explain magnetic ordering at high temperatures. One might expect that this requires a spin-dependent interaction, however, no such interaction of this strength is known. This is where direct exchange enters the stage. It is made up of multiple mechanisms, the first one (canonically) being Coulomb exchange. This idea goes back to Heisenberg. It is able to explain how ferromagnetic order comes about in a system of localized spins (e.g. spins of unpaired valence electrons). **More precise?** The most remarkable thing: the story starts with the (spin-independent!) Coulomb interaction between the electrons. In second quantization the Coulomb interaction reads

$$H_C = \frac{1}{2} \frac{e^2}{4\pi\epsilon_0} \int d^3r_1 \int d^3r_2 \sum_{\sigma_1, \sigma_2} \Psi_{\sigma_1}^\dagger(\vec{r}_1) \Psi_{\sigma_2}^\dagger(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \Psi_{\sigma_2}(\vec{r}_2) \Psi_{\sigma_1}(\vec{r}_1). \quad (1)$$

$\Psi_{\sigma_1}^\dagger(\vec{r}_1)$ is a field operator that creates an electron with spin σ_1 at position $\vec{r}_1$. These are now expanded in orthonormal wavefunctions localized around the ions (Wannier functions) $\phi_{\vec{R},m}(\vec{r})$ and spinors χ_σ :

$$\Psi_{\sigma_1}^\dagger(\vec{r}_1) = \sum_{\vec{R},m} a_{\vec{R}m\sigma_1}^\dagger \phi_{\vec{R}m}(\vec{r}_1) \chi_{\sigma_1}. \quad (2)$$

The sum over $\vec{R}$ goes over the positions of all Ions. m counts all angular degrees of freedom, but not the spin. $a_{\vec{R},m,\sigma_1}^\dagger$ is a fermionic creation operator. One finds

$$\begin{aligned} H_C &= \frac{1}{2} \frac{e^2}{4\pi\epsilon_0} \sum_{\substack{\vec{R}^{(1)} \dots \vec{R}^{(4)} \\ m^{(1)} \dots m^{(4)}}} \int d^3r_1 \int d^3r_2 \phi_{\vec{R}^{(1)}m^{(1)}}^*(\vec{r}_1) \phi_{\vec{R}^{(2)}m^{(2)}}^*(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \phi_{\vec{R}^{(3)}m^{(3)}}(\vec{r}_2) \phi_{\vec{R}^{(4)}m^{(4)}}(\vec{r}_1) \\ &\times \sum_{\sigma_1, \sigma_2} \chi_{\sigma_1}^\dagger \chi_{\sigma_2}^\dagger \chi_{\sigma_2} \chi_{\sigma_1} a_{\vec{R}^{(1)}m^{(1)}\sigma_1}^\dagger a_{\vec{R}^{(2)}m^{(2)}\sigma_2}^\dagger a_{\vec{R}^{(3)}m^{(3)}\sigma_2} a_{\vec{R}^{(4)}m^{(4)}\sigma_1}. \end{aligned} \quad (3)$$

The expression is simplified by evaluating the spinor product.

$$\begin{aligned} H_C &= \frac{1}{2} \frac{e^2}{4\pi\epsilon_0} \sum_{\substack{\vec{R}^{(1)} \dots \vec{R}^{(4)} \\ m^{(1)} \dots m^{(4)} \\ \sigma_1, \sigma_2}} \left\langle \vec{R}^{(1)}m^{(1)}, \vec{R}^{(2)}m^{(2)} \left| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right| \vec{R}^{(3)}m^{(3)}, \vec{R}^{(4)}m^{(4)} \right\rangle \\ &\times a_{\vec{R}^{(1)}m^{(1)}\sigma_1}^\dagger a_{\vec{R}^{(2)}m^{(2)}\sigma_2}^\dagger a_{\vec{R}^{(3)}m^{(3)}\sigma_2} a_{\vec{R}^{(4)}m^{(4)}\sigma_1} \end{aligned} \quad (4)$$

It is worthwhile to take a few detours in order to explore different effects of the Coulomb interaction, increasing the complexity in the mean-time.

on-site, single orbital: Hubbard U-term

First, focus on the on-site effects $\vec{R}^{(1)} = \vec{R}^{(2)} = \vec{R}^{(3)} = \vec{R}^{(4)}$, dropping the position quantum numbers. The easiest case is that of a single orbital per site, $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} \equiv m$.

$$\Rightarrow H_C = \frac{1}{2} \sum_{\vec{R}} \sum_{\sigma_1 \sigma_2} \left\langle \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right\rangle_{m,m} a_{\sigma_1}^\dagger a_{\sigma_2}^\dagger a_{\sigma_2} a_{\sigma_1} \quad (5)$$

The position indices of the ladder operators have been suppressed.

$$\sum_{\sigma_1 \sigma_2} a_{\sigma_1}^\dagger a_{\sigma_2}^\dagger a_{\sigma_2} a_{\sigma_1} = a_{\downarrow}^\dagger a_{\uparrow}^\dagger a_{\uparrow} a_{\downarrow} + a_{\uparrow}^\dagger a_{\downarrow}^\dagger a_{\downarrow} a_{\uparrow} = 2a_{\downarrow}^\dagger a_{\uparrow}^\dagger a_{\uparrow} a_{\downarrow} \quad (6)$$

Two out of the four terms ($\sigma_1 = \sigma_2 = \uparrow$ and $\sigma_1 = \sigma_2 = \downarrow$) are identically zero, since $(a_{\downarrow/\uparrow})^2 = 0$. In other words: due to Pauli principle the spin alignment is already fixed. So no exchange interaction can arise here. The technical reason is that we assumed a single orbital on a single site (2 states), leaving the electrons no other choice. We'll have to loosen the approximations to see exchange play out. Let's close with one last observation regarding the one-site, single orbital story;

$$\Rightarrow H_C = \sum_{\vec{R}} U a_{\downarrow}^\dagger a_{\uparrow}^\dagger a_{\uparrow} a_{\downarrow}, \quad U = \frac{e^2}{4\pi\epsilon_0} \int d^3 r_1 \int d^3 r_2 |\phi(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi(\vec{r}_2)|^2 \quad (7)$$

This gives the Hubbard U-term. It's interpretation is pretty straight forward: electrons occupying the same orbital have increased Coulomb energy, given by U . This presents a convenient way to include Coulomb interaction in tight-binding models. It is only an approximation, since all other terms (multiple sites, different orbitals) have been ignored so far. It is also clear that this contribution is the largest off all effects arising from the Coulomb interaction as the overlap is biggest here.

on-site, multiple orbitals: Hund's rule

The first step of complexification is to allow multiple orbitals at every ion, thereby reintroducing the $m^{(i)}$ quantum numbers. If one treats the Coulomb interaction perturbatively, in first order the creation/ annihilation operators have to be grouped into pairs in order to produce non-vanishing contributions. **I am not satisfied with this.** There are two possibilities: $m^{(1)} = m^{(3)}$, $m^{(2)} = m^{(4)}$ and $m^{(1)} = m^{(4)}$, $m^{(2)} = m^{(3)}$.

$$\begin{aligned} K_{m^{(1)}m^{(2)}} &:= \frac{e^2}{4\pi\epsilon_0} \left\langle m^{(1)}, m^{(2)} \left| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right| m^{(2)}, m^{(1)} \right\rangle \\ &= \frac{e^2}{4\pi\epsilon_0} \int d^3 r_1 \int d^3 r_2 |\phi_{m^{(1)}}(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi_{m^{(2)}}(\vec{r}_2)|^2 \end{aligned} \quad (8)$$

$$\begin{aligned} J_{m^{(1)}m^{(2)}} &:= \frac{e^2}{4\pi\epsilon_0} \left\langle m^{(1)}, m^{(2)} \left| \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right| m^{(1)}, m^{(2)} \right\rangle \\ &= \frac{e^2}{4\pi\epsilon_0} \int d^3 r_1 \int d^3 r_2 \phi_{m^{(1)}}^*(\vec{r}_1) \phi_{m^{(2)}}^*(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \phi_{m^{(1)}}(\vec{r}_2) \phi_{m^{(2)}}(\vec{r}_1) \end{aligned} \quad (9)$$

$$\begin{aligned} \Rightarrow H_C &\approx \frac{1}{2} \sum_{\vec{R}} \sum_{\substack{m^{(1)}, m^{(2)} \\ \sigma_1, \sigma_2}} \left(K_{m^{(1)}m^{(2)}} a_{m^{(1)}\sigma_1}^\dagger a_{m^{(2)}\sigma_2}^\dagger a_{m^{(2)}\sigma_2} a_{m^{(1)}\sigma_1} \right. \\ &\quad \left. + J_{m^{(1)}m^{(2)}} a_{m^{(1)}\sigma_1}^\dagger a_{m^{(2)}\sigma_2}^\dagger a_{m^{(1)}\sigma_2} a_{m^{(2)}\sigma_1} \right) \end{aligned} \quad (10)$$

The position indices have been suppressed! The exchange term may be reexpressed as

$$\begin{aligned} - \sum_{\vec{R}} \sum_{m^{(1)}, m^{(2)}} J_{m^{(1)}m^{(2)}} &\left(\left(a_{m^{(1)}\uparrow}^\dagger a_{m^{(1)}\uparrow} - a_{m^{(1)}\downarrow}^\dagger a_{m^{(1)}\downarrow} \right) \left(a_{m^{(2)}\uparrow}^\dagger a_{m^{(2)}\uparrow} - a_{m^{(2)}\downarrow}^\dagger a_{m^{(2)}\downarrow} \right) \right. \\ &\quad + a_{m^{(1)}\uparrow}^\dagger a_{m^{(1)}\downarrow} a_{m^{(2)}\downarrow}^\dagger a_{m^{(2)}\uparrow} + a_{m^{(1)}\downarrow}^\dagger a_{m^{(1)}\uparrow} a_{m^{(2)}\uparrow}^\dagger a_{m^{(2)}\downarrow} \\ &\quad \left. + a_{m^{(1)}\uparrow}^\dagger a_{m^{(1)}\uparrow} a_{m^{(2)}\downarrow}^\dagger a_{m^{(2)}\downarrow} + a_{m^{(1)}\downarrow}^\dagger a_{m^{(1)}\downarrow} a_{m^{(2)}\uparrow}^\dagger a_{m^{(2)}\uparrow} \right). \end{aligned} \quad (11)$$

Technically there are terms missing from the Fermi-algebra. However, these are only bilinear and thus act as a single-particle potential and are not relevant here. With the mapping/ substitution

$$n_m = a_{m\uparrow}^\dagger a_{m\uparrow} + a_{m\downarrow}^\dagger a_{m\downarrow}, \quad s_{mz} = \frac{1}{2} \left(a_{m\uparrow}^\dagger a_{m\uparrow} - a_{m\downarrow}^\dagger a_{m\downarrow} \right), \quad s_{m+} = a_{m\uparrow}^\dagger a_{m\downarrow} \quad (12)$$

one ends up with

$$H_C = \frac{1}{2} \sum_{\vec{R}} \sum_{m^{(1)}, m^{(2)}} \left(K_{m^{(1)} m^{(2)}} - \frac{1}{2} J_{m^{(1)} m^{(2)}} \right) n_{m^{(1)}} n_{m^{(2)}} - 2 J_{m^{(1)} m^{(2)}} \vec{s}_{m^{(1)}} \cdot \vec{s}_{m^{(2)}}. \quad (13)$$

The first term is the on-site Coulomb interaction between electrons in different orbitals on the same site. Additionally, $K_{m^{(1)} m^{(2)}} - \frac{1}{2} J_{m^{(1)} m^{(2)}} > 0$, meaning the Coulomb interaction remains repulsive. Returning to the exchange term; using

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \int \frac{d^3 k}{(2\pi)^3} \frac{4\pi}{k^2} e^{i\vec{k}(\vec{r}_1 - \vec{r}_2)} \quad (14)$$

$$\Rightarrow J_{m^{(1)} m^{(2)}} = \frac{e^2}{\varepsilon_0} \int \frac{d^3 k}{(2\pi)^3} \frac{1}{k^2} \underbrace{\int d^3 r_1 \phi_{m^{(1)}}^*(\vec{r}_1) \phi_{m^{(2)}}(\vec{r}_1) e^{i\vec{k}\vec{r}_1}}_{I(\vec{r})} \underbrace{\int d^3 r_2 \phi_{m^{(1)}}(\vec{r}_2) \phi_{m^{(2)}}^*(\vec{r}_2) e^{-i\vec{k}\vec{r}_2}}_{I^*(\vec{r})}. \quad (15)$$

Hence, $J_{m^{(1)} m^{(2)}}$ is positive. Therefore the interaction is minimized by ferromagnetic allignment. This is the derivation of Hund's first rule: the ground state of a partially filled shell is the one that maximizes $\vec{s}_{\text{tot}}$.

different site, single orbital: Coulomb exchange

Assuming again singular orbitals $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} \equiv m$, whilst reinstating the position quantum numbers the calculation is symmetrical to that in the last subchapter.

$$\Rightarrow H_C = \frac{1}{2} \sum_{\vec{R}^{(1)}, \vec{R}^{(2)}} \left(K_{\vec{R}^{(1)} \vec{R}^{(2)}} - \frac{1}{2} J_{\vec{R}^{(1)} \vec{R}^{(2)}} \right) n_{\vec{R}^{(1)}} n_{\vec{R}^{(2)}} - 2 J_{\vec{R}^{(1)} \vec{R}^{(2)}} \vec{s}_{\vec{R}^{(1)}} \cdot \vec{s}_{\vec{R}^{(2)}} \quad (16)$$

$$K_{\vec{R}^{(1)} \vec{R}^{(2)}} = \frac{e^2}{4\pi\varepsilon_0} \int d^3 r_1 \int d^3 r_2 |\phi_{\vec{R}^{(1)}}(\vec{r}_1)|^2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\phi_{\vec{R}^{(2)}}(\vec{r}_2)|^2 \quad (17)$$

$$J_{\vec{R}^{(1)} \vec{R}^{(2)}} = \frac{e^2}{4\pi\varepsilon_0} \int d^3 r_1 \int d^3 r_2 \phi_{\vec{R}^{(1)}}^*(\vec{r}_1) \phi_{\vec{R}^{(2)}}^*(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \phi_{\vec{R}^{(1)}}(\vec{r}_2) \phi_{\vec{R}^{(2)}}(\vec{r}_1) \quad (18)$$

As in the last chapter $J_{\vec{R}^{(1)} \vec{R}^{(2)}} > 0$, so again the interaction is ferromagnetic. The interpretation is the same as for the previous subchapter; electrons in a triplet state avoid each other in real space due to Pauli-principle ("exchange hole"), thereby reducing their Coulomb energy.

In terms of explaining magnetic ordering this obviously can't be the end of the story, since we haven't found a way to explain anti-ferromagnetic order yet.

2. Partons

2.1 SU(2) Slave Fermions

The method described here to represent spins in terms of Fermions was first employed by Abrikosov in 1965 and later reformulated in terms of gauge theory by Affleck in 1988. The idea (for spin 1/2!) is to introduce two new flavours of fermions: $c_{i\uparrow}^\dagger, c_{i\downarrow}^\dagger$ on each site. The transformation reads

$$\vec{S}_i = \frac{1}{2} \sum_{\alpha, \beta} c_{i\alpha}^\dagger \vec{\sigma}_{\alpha\beta} c_{i\beta}, \quad (19)$$

where $\vec{\sigma}$ is the vector of Pauli matrices.

This transformation is quite natural, as it has been found even before Abrikosov, just the other way around. It is known that the large- U limit of the Hubbard model at half-filling gives a antiferromagnetic Heisenberg model. In the perturbation theory one finds the exact representation of Spins in terms of fermionic ladder operators given above.

2.1.1 Hilbert space and un-physical states

The mapping 19 is exact on the operator level, meaning, that the eigenspaces of S_i^2 and S_i^α remain unchanged.

The local Hilbert space, however, gets enlarged. We drop the site-indices from now on. The spin-1/2 Hilbert space is of course a two-dimensional Hilbert space

$$\mathcal{H}_{S=\frac{1}{2}} = \text{span}(|\uparrow\rangle, |\downarrow\rangle) \quad (20)$$

The two-fermion Hilbert space on the other hand is four-dimensional

$$\mathcal{H}_{\text{fermions}} = \text{span}(|00\rangle, |10\rangle, |01\rangle, |11\rangle). \quad (21)$$

The idea of 19 is to identify $\mathcal{H}_{S=\frac{1}{2}}$ with the one-particle sector of $\mathcal{H}_{\text{fermions}}$:

$$(n_{i\uparrow} + n_{i\downarrow}) |\psi\rangle \stackrel{!}{=} |\psi\rangle \quad \forall i. \quad (22)$$

This leaves us with two un-physical states: $|00\rangle$ and $|11\rangle$.

All spin-operators S^α act trivially on these, which is why everything goes through smoothly on the operator level. Therefore, as long as the Hamiltonian only consists of spin operators, the un-physical states do not influence any results (apart from normalization, maybe).

This can not be guaranteed anymore when treating the system perturbatively using for example a mean-field theory. For this reason we introduce the constraint

$$\langle \psi | n_{i\downarrow} + n_{i\uparrow} | \psi \rangle = 1 \quad \forall i. \quad (23)$$

which is a constraint on the state $|\psi\rangle$. Acknowledge, that this constraint is weaker than $n_i = 1 \quad \forall i$, for example

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad \Rightarrow \quad \langle \psi | n_\uparrow + n_\downarrow | \psi \rangle = 1 \quad \text{but} \quad (n_{i\uparrow} + n_{i\downarrow}) |\psi\rangle = \sqrt{2} |11\rangle \neq |\psi\rangle. \quad (24)$$

Obviously all normalized states that fulfill 22 also comply with 23.

2.1.2 global SU(2) symmetrie

Many important models, e.g. exchange models, have spin rotation symmetrie. The symmetrie group of spin-rotations is SU(2). Rewriting 19 in Matrix notation (for $S=1/2$, without position indices):

$$\vec{S} = \frac{1}{2} \begin{pmatrix} c_\uparrow^\dagger & c_\downarrow^\dagger \\ c_\uparrow & c_\downarrow \end{pmatrix} \vec{\sigma} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \quad (25)$$

the spin rotation can be represented by $\vec{\sigma} \mapsto V^{-1} \vec{\sigma} V$, with V an element of SU(2). The task at hand is to determine, how the spin rotation acts on the ladder operators. One may reinterpret

$$\vec{S} \rightarrow \frac{1}{2} \begin{pmatrix} c_{\uparrow}^{\dagger} & c_{\downarrow}^{\dagger} \end{pmatrix} V^{-1} \vec{\sigma} V \begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix} \quad (26)$$

as

$$\begin{pmatrix} c_{\uparrow}^{\dagger} & c_{\downarrow}^{\dagger} \end{pmatrix} \rightarrow \begin{pmatrix} c_{\uparrow}^{\dagger} & c_{\downarrow}^{\dagger} \end{pmatrix} V^{-1} \quad \begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix} \rightarrow V \begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix} \quad \vec{\sigma} \rightarrow \vec{\sigma}. \quad (27)$$

$c_{\downarrow}, c_{\uparrow}$ transform as a SU(2) doublet under left multiplication by an SU(2) matrix. To construct the conjugate doublet, which transforms in the same manner, consider

$$\begin{pmatrix} c_{\uparrow}^{\dagger} \\ c_{\downarrow}^{\dagger} \end{pmatrix} \rightarrow \left[\begin{pmatrix} c_{\uparrow}^{\dagger} & c_{\downarrow}^{\dagger} \end{pmatrix} V^{-1} \right]^T = (V^{-1})^T \begin{pmatrix} c_{\uparrow}^{\dagger} \\ c_{\downarrow}^{\dagger} \end{pmatrix} = (V^{-1})^T \sigma_3 \sigma_3 \begin{pmatrix} c_{\uparrow}^{\dagger} \\ c_{\downarrow}^{\dagger} \end{pmatrix} \quad (28)$$

To recover V^{-1} , multiply both sides by σ_3 from the left and use $V^{-1} = V^H = \sigma_3 V^T \sigma_3$ (σ_3 represents the complex conjugation in SU(2) **better formulation?**).

$$\Rightarrow \sigma_3 \begin{pmatrix} c_{\uparrow}^{\dagger} \\ c_{\downarrow}^{\dagger} \end{pmatrix} \rightarrow V \sigma_3 \begin{pmatrix} c_{\uparrow}^{\dagger} \\ c_{\downarrow}^{\dagger} \end{pmatrix} \quad (29)$$

Hence, the two conjugate doublets are

$$\begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} c_{\uparrow}^{\dagger} \\ -c_{\downarrow}^{\dagger} \end{pmatrix} \quad (30)$$

It is convention two write these two doublets in a matrix:

$$\psi = \begin{pmatrix} c_{\uparrow} & c_{\downarrow} \\ c_{\downarrow}^{\dagger} & -c_{\uparrow}^{\dagger} \end{pmatrix}, \quad (31)$$

the spin rotation is then $\psi \mapsto \psi V$. Hence, the combination $\psi_i \psi_j^{\dagger}$ is manifestly invariant. This reveals two rotationally invariant operators χ_{ij} and η_{ij} :

$$\psi \psi^{\dagger} = \begin{pmatrix} c_{i\uparrow}^{\dagger} c_{j\uparrow}^{\dagger} + c_{i\downarrow}^{\dagger} c_{j\downarrow}^{\dagger} & c_{i\uparrow}^{\dagger} c_{j\downarrow} - c_{i\downarrow}^{\dagger} c_{j\uparrow} \\ c_{i\downarrow}^{\dagger} c_{j\uparrow} - c_{i\uparrow}^{\dagger} c_{j\downarrow} & c_{i\downarrow}^{\dagger} c_{j\downarrow} + c_{i\uparrow}^{\dagger} c_{j\uparrow} \end{pmatrix} \equiv \begin{pmatrix} -\chi_{ij}^{\dagger} & -\eta_{ij}^{\dagger} \\ -\eta_{ij} & \chi_{ij} \end{pmatrix} \quad (32)$$

The spint operators can be expressed in manifestly rotationally-invariant form:

$$S_i^{\alpha} = \frac{1}{2} \text{Tr} \left[\psi_i^{\dagger} \psi_i (\sigma)^T \right] \quad (33)$$

2.1.3 local SU(2) symmetry

Consider

$$S_+ |\downarrow\rangle = |\uparrow\rangle, \quad (34)$$

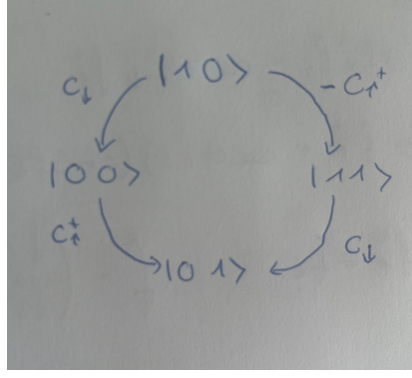
which is an operation on $\mathcal{H}_{S=\frac{1}{2}}$. When translating this into $\mathcal{H}_{\text{fermions}}$:

$$S_+ = \frac{1}{2} (S_x + iS_y) \rightarrow c_{\uparrow}^{\dagger} c_{\downarrow}, \quad (35)$$

one uncovers a redundancy in the description:

$$c_{\uparrow}^{\dagger} c_{\downarrow} = \sin^2(\theta) c_{\uparrow}^{\dagger} c_{\downarrow} + \cos^2(\theta) c_{\uparrow}^{\dagger} c_{\downarrow} = \sin^2(\theta) c_{\uparrow}^{\dagger} c_{\downarrow} - \cos^2(\theta) c_{\downarrow} c_{\uparrow}^{\dagger}. \quad (36)$$

On the level of local hilbert spaces, this can be understood as different walks connecting the same initial and final states (hence, an artifact of the enlargement).



The physical interpretation is that $c_\downarrow$ and $-c_\uparrow^\dagger$ have the same action: they decrease S_z by unit. Suddenly, we have a gauge-structure at our hands. The next questions come naturally: what are general gauge transformations and what are gauge invariants? We combine the four operators into two doublets:

$$\begin{pmatrix} c_\downarrow \\ -c_\uparrow^\dagger \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} c_\uparrow \\ c_\downarrow^\dagger \end{pmatrix}, \quad (37)$$

which transform by left multiplication by a matrix (the gauge transformation). These doublets again form 36, From the representation of the spin-operators it is obvious that the gauge transformations have to be from $SU(2)$.

3. Mechanics of Continua

3.1 Kinematics

3.1.1 Notation and Coordinates

I will skip over the very technical mathematical description of the basic objects (like bodies and points) and skip ahead to the interesting stuff.

It is convention to use large latin letters for quantities in the reference configuration and small latin letters for quantities of a particular configuration. One might also refer to the reference configuration as the initial configuration.

For example: the position of a particle in the reference configuration is called X , while it is called x in a particular configuration.

This notation gets inherited by all objects in the theory through the variables they depend on.

3.1.2 Eulerian and Lagrangian description

When describing how the configuration of a body changes over time it can be advantageous to describe different processes from the points of view of different observers. It comes as no surprise that the choice of description does not change the physics but is merely a matter of convenience. There are two canonical descriptions to choose from;

In the **Eulerian description** (or spatial description) $X(x, t)$ is a function of t and x .

The intuitive picture is that of an observer that observes a certain point x in space. Like a hiker resting next to stream, observing it as it flows by. If they encounter particle P at time t (at their point of observation x), they can infer the point $X_P(x, t)$, where P would have been in the reference configuration.

In this sense x is the only free variable besides t . Therefore, quantities in the Eulerian picture are always denoted by small latin letters.

In the **Lagrangian description** (or material description) $x(X, t)$ is a function of t and X .

In this case the observer follows a particle (or a collection of particles, called a region) as it evolves over time. Imagine the hiker built a raft and rides it down the aforementioned stream (assuming the rafts velocity matches that of the stream). Knowing the point X where they started rafting they can reason where they will end up at time t ; the point being $x(X, t)$.

Hence, here X and t are the free coordinates. All material quantities are denoted by capital greek letters.

3.1.3 Trajectories and Displacement Field

The object connecting the aforementioned descriptions is the **trajectorie** $\chi(X, t)$:

$$x = \chi(X, t) \quad \Leftrightarrow \quad X = \chi^{-1}(x, t). \quad (38)$$

It maps out where a observed particle that started at X will be at time t . Obviously, χ itself is part of the Lagrangian description (χ^{-1} belongs to the Eulerian description). A function $f(x, t)$ in Eulerian description can be transformed according to:

$$f(x, t) \mapsto F(X, t) = f(\chi(X, t), t), \quad (39)$$

where F is now in the Lagrangian description. Though f and F are conceptually equivalent they are, mathematically speaking, two different objects.

An example for such a function is the **displacement field**:

$$\begin{aligned} U(X, t) &= X - x(X, t) \\ u(x, t) &= X(x, t) - x \end{aligned} \quad (40)$$

$$U(X, t) \mapsto u(x, t) = U(\chi^{-1}(x, t), t) \quad (41)$$

The displacement field describes how much a particle is displaced compared to the reference configuration.

3.1.4 local changes and Deformation Gradient

Part II

main part

4. Triangular Lattice

4.1 Construction of Hamiltonian

rename this section to something like model?

4.1.1 Construction of the Hamiltonian

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \quad (42)$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f_{i\alpha}^\dagger \frac{\vec{\sigma}}{2} f_{i\beta} \quad (43)$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f_{i\alpha}^\dagger f_{i\beta} f_{j\gamma}^\dagger f_{j\delta} \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\gamma}\delta_{\beta\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2f_{i\alpha}^\dagger f_{i\beta} f_{j\beta}^\dagger f_{j\alpha} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second position: use $f_{j\beta}^\dagger f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f_{j\beta}^\dagger$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{j\beta} + 2f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \quad (44)$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f_{i\alpha}^\dagger f_{j\alpha} \quad (45)$$

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{2}n_i - \frac{1}{4}n_i n_j \quad (46)$$

Using the constraint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{4} \quad (47)$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij}\chi_{ji} + 4(\chi_{ij}\chi_{ji})^2 \quad (48)$$

The mean field decoupling works like

$$\chi_{ij}\chi_{ji} \approx \langle\chi_{ij}\rangle\chi_{ji} + \langle\chi_{ji}\rangle\chi_{ij} - |\langle\chi_{ij}\rangle|^2 \quad (49)$$

$$(\chi_{ij}\chi_{ji})^2 \approx 2|\langle\chi_{ij}\rangle|^2\langle\chi_{ij}\rangle\chi_{ji} + 2|\langle\chi_{ij}\rangle|^2\langle\chi_{ji}\rangle\chi_{ij} - 3|\langle\chi_{ij}\rangle|^4 \quad (50)$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2\langle\chi_{ij}\rangle\chi_{ji} + \text{h.c.} + 2|\langle\chi_{ij}\rangle|^2 + \frac{1}{4} \quad (51)$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j\right)^2 \approx -\langle\chi_{ij}\rangle\chi_{ji} + 8|\langle\chi_{ij}\rangle|^2\langle\chi_{ij}\rangle\chi_{ji} + \text{h.c.} + |\langle\chi_{ij}\rangle|^2 - 12|\langle\chi_{ij}\rangle|^4 + \frac{1}{16} \quad (52)$$

Maybe comment of the fact that the mean-field parameters know about the fermi sea and thereby about many states. In this sense the states do speak with each other. Bringing everything together:

$$\begin{aligned} \mathcal{H} \approx \sum_{\langle ij \rangle} J_{ij} & \left[\left(-1 - \kappa/2 + 4\kappa|\langle\chi_{ij}\rangle|^2 \right) \langle\chi_{ij}\rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\ & \left. + \left(2 + \kappa - 12\kappa|\langle\chi_{ij}\rangle|^2 \right) |\langle\chi_{ij}\rangle|^2 + \frac{1}{4} + \frac{\kappa}{16} \right] \end{aligned} \quad (53)$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij} (1 + \kappa/2), \quad \tilde{\kappa} = \kappa/(1 + \kappa/2) \quad (54)$$

I drop the constant:

$$\begin{aligned} \mathcal{H} \approx \sum_{\langle ij \rangle} \tilde{J}_{ij} & \left[- \left(1 - 4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2 \right) \langle\chi_{ij}\rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\ & \left. + 2 \left(1 - 6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2 \right) |\langle\chi_{ij}\rangle|^2 \right] \end{aligned} \quad (55)$$

Hence, the mean field produces the effective hopping amplitude t^{MF} :

$$t_{ji}^{\text{MF}} = -\tilde{J}_{ij}\langle\chi_{ij}\rangle \left(1 - 4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2 \right) = -J_{ij}\langle\chi_{ij}\rangle \left(1 + \kappa/2 - 4\kappa|\langle\chi_{ij}\rangle|^2 \right). \quad (56)$$

$$\Rightarrow \mathcal{H} \approx \sum_{\langle ij \rangle} t_{ji}^{\text{MF}} \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{const.} \quad (57)$$

This is pretty interesting: the hopping-amplitude $i \rightarrow j$ is directly proportional to the hopping-amplitude $j \rightarrow i$. Here are a few ideas how to make use/discuss this insight:

- Maybe one can also make the argument that we start with a "symmetric" model, i.e. a model without internal sense of direction, a feature that is now preserved in the Mean field Hamiltonian.
- Or maybe, even more sophisticated, one could make argument using structure of the spin-singlet states.
- Maybe one can use this fact to simplify the Mean-field equation?

4.1.2 Continuity equation and probability current

The probability (/particle) density on site k is

$$n_k = \sum_{\sigma} f_{k\sigma}^{\dagger} f_{k\sigma}. \quad (58)$$

Now I derive the corresponding probability current density:

$$\frac{\partial n_k}{\partial t} = -i \sum_{\sigma} \left([f_{k\sigma}^{\dagger}, H] f_{k\sigma} + f_{k\sigma}^{\dagger} [f_{k\sigma}, H] \right) \quad (59)$$

$$\begin{aligned} [f_{k\sigma}^{\dagger}, H] f_{k\sigma} &= - \sum_{\delta} t_{k+\delta; k} f_{k+\delta; \sigma}^{\dagger} f_{k\sigma} - \mu_k f_{k\sigma}^{\dagger} f_{k\sigma} \\ f_{k\sigma}^{\dagger} [f_{k\sigma}, H] &= \sum_{\delta} t_{k; k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta; \sigma} + \mu_k f_{k\sigma}^{\dagger} f_{k\sigma} \end{aligned} \quad (60)$$

$$\Rightarrow \frac{\partial n_k}{\partial t} = - \sum_{\delta} i \left(\sum_{\sigma} t_{k, k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta; \sigma} - t_{k+\delta, k} f_{k+\delta; \sigma}^{\dagger} f_{k\sigma} \right) \quad (61)$$

Define

$$J_{k \rightarrow k+\delta} \equiv i \left(\sum_{\sigma} t_{k, k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta; \sigma} - t_{k+\delta, k} f_{k+\delta; \sigma}^{\dagger} f_{k\sigma} \right) \quad (62)$$

$J_{k \rightarrow k+\delta}$ is the (directed!) current density along the bond $\vec{R}_k \rightarrow \vec{R}_{k+\delta}$. The discrete (/lattice) continuity equation reads

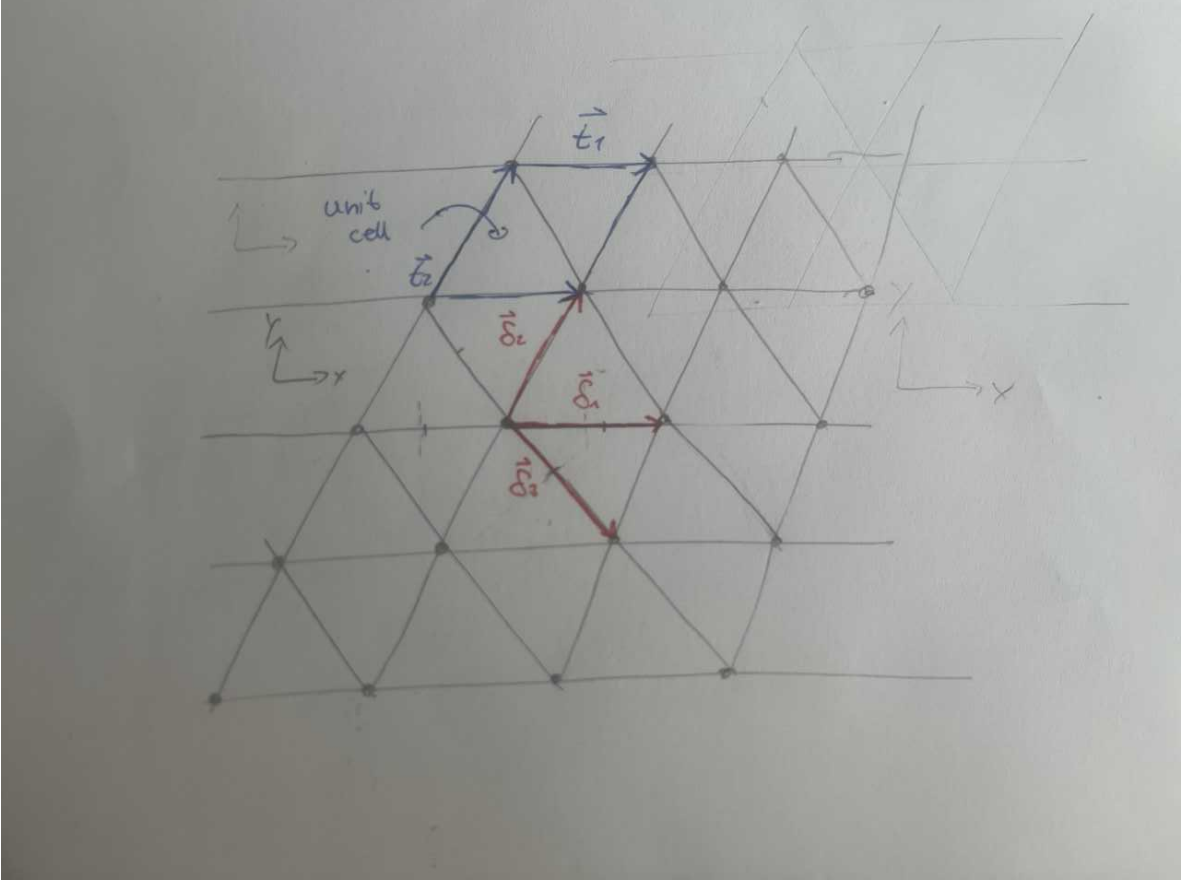
$$\frac{\partial n_k}{\partial t} = - \sum_{\delta} J_{k \rightarrow k+\delta}. \quad (63)$$

Something interesting happens when calculating $\langle J_{k \rightarrow k+\delta} \rangle$:

$$\begin{aligned} \langle J_{k \rightarrow k+\delta} \rangle &= i \left(t_{k, k+\delta} \left\langle \sum_{\sigma} f_{k\sigma}^{\dagger} f_{k+\delta; \sigma} \right\rangle - t_{k+\delta, k} \left\langle \sum_{\sigma} f_{k+\delta; \sigma}^{\dagger} f_{k\sigma} \right\rangle \right) \\ &= 2i (t_{k, k+\delta} \langle \chi_{k, k+\delta} \rangle - t_{k+\delta, k} \langle \chi_{k+\delta, k} \rangle) \\ &= -2i \tilde{J}_{k; k+\delta} \left(1 - 4\tilde{\kappa} |\langle \chi_{k, k+\delta} \rangle|^2 \right) \left(\langle \chi_{k+\delta; k} \rangle \langle \chi_{k; k+\delta} \rangle - \langle \chi_{k; k+\delta} \rangle \langle \chi_{k+\delta; k} \rangle \right) \\ &= 0. \end{aligned} \quad (64)$$

Apparently, the current on each bond vanishes in the ground state by construction. This still leaves the possibility that single states have non-zero probability currents. However, it remains to investigate how this is possible for e.g. states with a singular (topologically protected) edge mode. One would guess that this feature would lead to a non-zero current in the ground state.

4.2 0-0 and π - π States



This system is a simple triangular Bravais lattice with primitive translations

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}. \quad (65)$$

4.2.1 Pedestrian method

The tight-binding Hamiltonian is:

$$\begin{aligned} H &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f_{i\sigma}^\dagger f_{i+\delta\sigma} + \text{h.c.} \right) + \mu \sum_i \left(f_{i\uparrow}^\dagger f_{i\uparrow} + f_{i\downarrow}^\dagger f_{i\downarrow} - 1 \right) + \text{const.} \\ &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f_{i\sigma}^\dagger f_{i+\delta\sigma} + \text{h.c.} \right) + \mu \sum_{i,\sigma} f_{i\sigma}^\dagger f_{i\sigma} - \mu N + \text{const..} \end{aligned} \quad (66)$$

i runs over sites and δ over nearest neighbours of a generic site. I include the absolute shift for the sake of consistency. The factor 1/2 is also included for consistency. The constant is a product of mean field theory and reads

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) \right) |\langle \chi_{i,i+\delta} \rangle|^2 \quad (67)$$

The second step can be understood in at least two ways; a) each site has 6 neighbours, or b) there are bond per primitive unit cell 1 of which is completely inside the unit cell and contributes a 2 and the

other 4 shared between 2 unit cells thereby contributing a 1. All bonds are equivalent of course. The aforementioned six nearest neighbours lie at the (relative) positions

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \quad \vec{\delta}_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

and $-\vec{\delta}_1, -\vec{\delta}_2, -\vec{\delta}_3$. I will start with the 0-0 state. To advance, do a Fourier transform:

$$f_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k}\vec{R}_i} c_{k\sigma}. \quad (68)$$

$$\begin{aligned} \Rightarrow H &= \sum_{q,\sigma} \frac{t}{2} \left(c_{q\sigma}^\dagger c_{q\sigma} \left(\sum_{\delta} e^{i\vec{q}\vec{\delta}} + \mu \right) + \text{h.c.} \right) - \mu N \\ &= \sum_{q,\sigma} \left(t \sum_{\delta} \cos(\vec{q}\vec{\delta}) + \mu \right) c_{q\sigma}^\dagger c_{q\sigma} - \mu N \end{aligned} \quad (69)$$

Using first the symmetrie of the cosine and then the sum identity for the cosine the dispersion reads

$$\begin{aligned} \varepsilon(\vec{q}) &= 2t \left[\cos(q_x) + \cos\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \right] + \mu \\ &= 2t \left[\cos(q_x) + 2 \cos\left(\frac{q_x}{2}\right) \cos\left(\frac{\sqrt{3}q_y}{2}\right) \right] + \mu \end{aligned} \quad (70)$$

features:

- $\varepsilon(\vec{q}) \in [-3t, 6t] + \mu \Rightarrow \mu \in [-6t, 3t]$
- symmetries $q_x \rightarrow -q_x$ and $q_y \rightarrow -q_y$ (+ should have some rotational symmetrie and (thereby associated) inflections along other axes) **could check this**
- periodicity: 4π in q_x , $(4\pi)/\sqrt{3}$ in q_y

The mean field parameters are

$$\langle \chi_{ij} \rangle = \frac{1}{2} \langle f_{i\uparrow}^\dagger f_{j\uparrow} + f_{i\downarrow}^\dagger f_{j\downarrow} \rangle = \langle f_{i\uparrow}^\dagger f_{j\uparrow} \rangle \quad (71)$$

Simplify the expression by Fourier transform:

$$\begin{aligned} \langle \chi_{ij} \rangle &= \frac{1}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{k}\vec{R}_i)} \langle c_p^\dagger c_q \rangle \\ &= \frac{1}{N} \sum e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \end{aligned} \quad (72)$$

Take the thermodynamic limit $\sum_k \rightarrow (N\Omega)/(2\pi)^2 \int dk$ (Ω is the unit cell volume).

$$\Rightarrow \langle \chi_{ij} \rangle = \Omega \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \quad (73)$$

It looks like the model allows for different χ_{ij} in each direction $\vec{R}_j - \vec{R}_i$ (there are 3 of those)?

From this expression, recover the on-site particle number by setting $i = j$ and accounting for the factor 1/2 in the definition of χ_{ij} :

$$\langle N_i \rangle = 2\Omega \int \frac{d^2 q}{(2\pi)^2} f(\varepsilon(\vec{q})). \quad (74)$$

By construction the particle number is uniform.

Some elaborations on the integral domain follow. The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (75)$$

I do a basis transformation onto those axes defined by

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \quad (76)$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (77)$$

$$\Rightarrow \det J = -2/\sqrt{3} \quad (78)$$

$$\Rightarrow \varepsilon(\vec{k}) = 2t (\cos(k_x) + \cos(k_y) + \cos(k_x - k_y)) + \mu \quad (79)$$

The first Brillouin zone is $\{k_x, k_y\} \in [-\pi, \pi]^2$. The unit cell volume is $\Omega = \sqrt{3}/2$. Consistency check: $|\det(J)| = \Omega^{-1}$, since the substitution simply transforms the unit cell to a square of unit volume. So the integrals that have to be evaluated are of the form

$$\langle N_i \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} f(\varepsilon(\vec{k}), \mu), \quad (80)$$

$$\langle \chi_{ij} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, (1/\sqrt{3})(2k_x - k_y) \right) \cdot (\vec{R}_j - \vec{R}_i) \right] f(\varepsilon(\vec{k}), \mu) \quad (81)$$

Check that under a particle hole transformation χ changes sign, sign of t is doing just that and this brings the π -flux phase.

The condition is $\langle N_i \rangle \stackrel{!}{=} 1 \forall i$. By setting $t = -1$, the chemical potential is numerically determined in units of t using Mathematica. I choose $t = -1$ because I anticipate t to be negative for a positive χ . (This may not be true depending on κ and χ , but as long as it is self-consistent everything is fine.) For the numerical integration I used a Fermi-function with a small temperature ($\beta = 200t$) instead of a step-function, since the ladder is not analytical and causes problems for the numerical algorithm. The result is $\mu = -0.834737 t$. The bond-parameters are calculated the same way, the result is $\langle \chi_{ij} \rangle = 0.164713$.

Now, it is possible to express μ only in units of J . Use the formula

$$t_{ij} = -J \langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2). \quad (82)$$

Set $\kappa = 10$:

$$t_{ij} = -0.8095 J \quad (83)$$

$$\mu = -0.67574 J \quad (84)$$

$$\langle \chi_{ij} \rangle = 0.164713 \quad (85)$$

The energy density (that is also the free energy density at $T = 0$) is

$$\frac{\langle H \rangle}{N} = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon(\vec{k}) f(\varepsilon(\vec{k})) - \mu + 6\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i, i+\delta} \rangle|^2 \right) |\langle \chi_{i, i+\delta} \rangle|^2 = -0.888304 J \quad (86)$$

Observation; the chemical potential drops out of the Free energy.

One can also directly calculate results for the state with all fluxes π by performing a particle-hole-transformation:

$$t \rightarrow -t \quad (87)$$

$$\mu \rightarrow -\mu \quad (88)$$

$$\Rightarrow \varepsilon(\vec{k}) \rightarrow -\varepsilon(\vec{k}) \quad (88)$$

$$\Rightarrow f(\varepsilon(\vec{k})) \rightarrow 1 - f(\varepsilon(\vec{k})) \quad (89)$$

It can be shown that

$$\chi \rightarrow -\chi. \quad (90)$$

The result is:

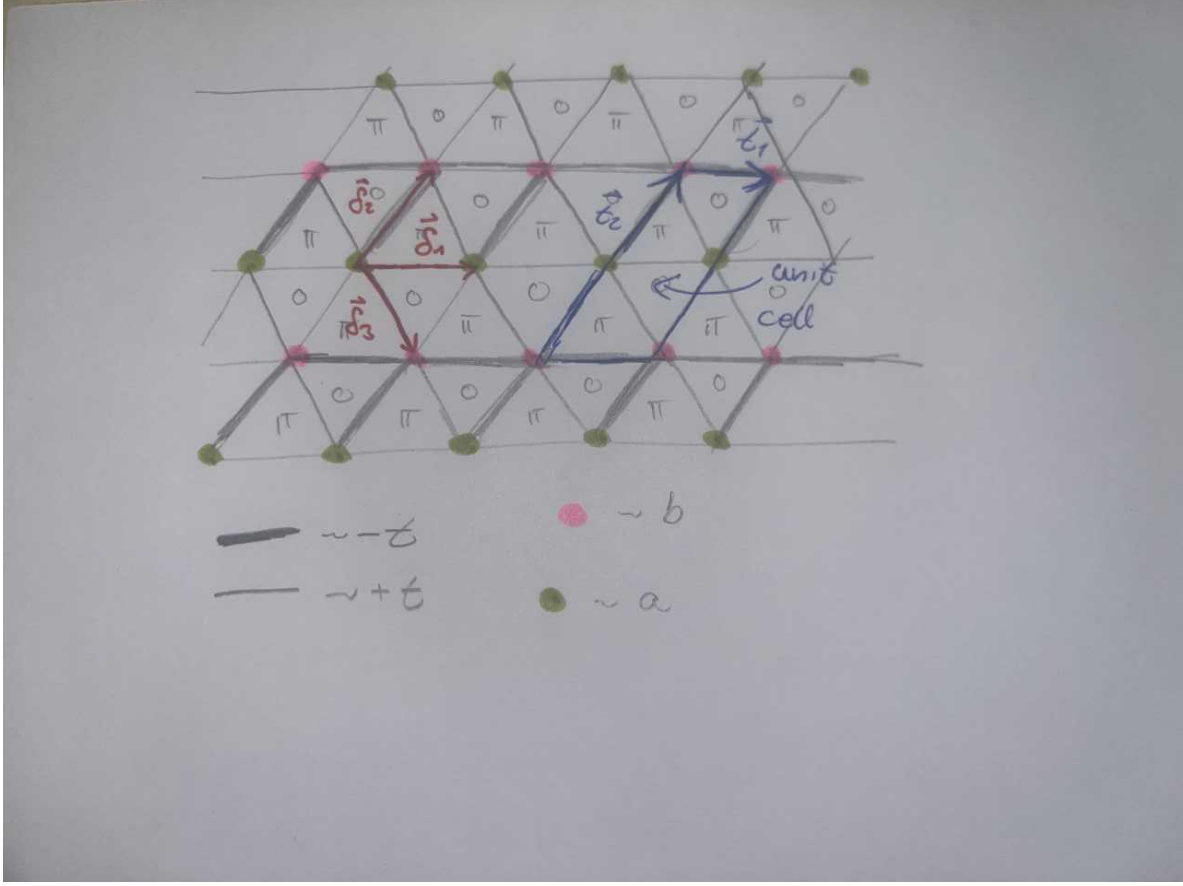
$$\frac{\langle H \rangle_\pi}{N} = -0.888302 J \quad (91)$$

4.2.2 Pedro's method

The longer explanation for this method is in the next section. Here I will only say that the free energy is the same except for γ , which gets replaced by a factor ϕ (as mentioned above the chemical potential does not contribute to the free energy):

4.3 0- π and π -0 States

4.3.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitive translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad (92)$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij} , and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2} \sum_{\langle i,j \rangle, \sigma} \left(t a_{i\sigma}^\dagger a_{j\sigma} - t b_{i\sigma}^\dagger b_{j\sigma} + t_{ij}^{(ab)} a_{i\sigma}^\dagger b_{j\sigma} + \text{h.c.} \right) + \mu_a \sum_{i,\sigma} \left(a_{i\sigma}^\dagger a_{i\sigma} - 1 \right) + \mu_b \sum_{i,\sigma} \left(b_{i\sigma}^\dagger b_{i\sigma} - 1 \right) + \text{const.} \quad (93)$$

$$\begin{aligned} H = \frac{t}{2} \sum_{i\sigma} & \left(a_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + a_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \\ & - b_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - b_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_3;\sigma} + b_{i\sigma}^\dagger a_{i-\delta_3;\sigma} \\ & \left. + \mu_a a_{i\sigma}^\dagger a_{i\sigma} + \mu_b b_{i\sigma}^\dagger b_{i\sigma} \right) - \frac{N(\mu_a + \mu_b)}{2a} \end{aligned} \quad (94)$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6|\langle \chi_{i,i+\delta} \rangle|^2\right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N\tilde{J} \left(1 - 6\tilde{\kappa}|\langle \chi_{i,i+\delta} \rangle|^2\right) |\langle \chi_{i,i+\delta} \rangle|^2 \quad (95)$$

This is not quite right yet. The Fourier transformation has to be modified bc there are only N/2 of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}} \sum_k e^{i\vec{k}\vec{R}_i} a_{k\sigma}, \quad (96)$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t \cos(\vec{q}\vec{\delta}_1) + \mu_a & 2t \cos(\vec{q}\vec{\delta}_3) - 2it \sin(\vec{q}\vec{\delta}_2) \\ 2t \cos(\vec{q}\vec{\delta}_3) + 2it \sin(\vec{q}\vec{\delta}_2) & -2t \cos(\vec{q}\vec{\delta}_1) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \quad (97)$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (98)$$

The suitable transformation is

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1 & \sqrt{3} \\ m1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \quad (99)$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (100)$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \quad (101)$$

This yields

$$\begin{aligned} \cos(\vec{q}\vec{\delta}_1) &\rightarrow \cos(k_y) \\ \sin(\vec{q}\vec{\delta}_2) &\rightarrow \sin(k_x/2) \end{aligned} \quad (102)$$

$$\cos(\vec{q}\vec{\delta}_3) \rightarrow \cos(k_y - k_x/2) \quad (103)$$

$$\varepsilon(\vec{k})_{\pm} = \mu_a + \mu_b \pm \sqrt{(\mu_a - \mu_b)^2 + 2t \cos(k_y)(\mu_a - \mu_b) + 4t^2 \cos^2(k_y) + 4t^2 \cos^2(k_y - k_x/2) + 4t^2 \sin^2(k_x/2)} \quad (104)$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-i charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_{\pm} = \pm 2|t| \sqrt{(\mu/(2t) + \cos(k_y))^2 + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \quad (105)$$

It is correct that ε only depends on the absolute value of t, it looks like this would completely neutralise the particle-hole transformation. The periodicity is a bit weird here, since the expression is not periodic in 2π in every direction as one would expect. I am pretty sure this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brillouin zone got cut in half in one direction.

represent hamiltonian as

$$H = \sum_{q,\sigma} (a_{q\sigma}^\dagger \quad b_{q\sigma}^\dagger) \vec{h} \cdot \vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \quad (106)$$

$$\vec{h} = \begin{pmatrix} 2t \cos(k_y - k_x/2) \\ 2t \sin(k_x/2) \\ 2t \cos(k_y) + \mu \end{pmatrix} \quad (107)$$

Then $\varepsilon(\vec{q})_\pm = \pm |\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t \cos(k_y) + \mu)}} \begin{pmatrix} \varepsilon_+ + 2t \cos(k_y) + \mu & -2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) \\ 2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) & \varepsilon_+ + 2t \cos(k_y) + \mu \end{pmatrix} \quad (108)$$

is the transformation matrix U . It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned} \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &\stackrel{\text{FT}}{=} \frac{2}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a_p^\dagger b_q \rangle}_{\propto \delta_{q,p}} \\ &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \langle a_q^\dagger b_q \rangle \\ &\stackrel{\text{BT}}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} (u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-)) \end{aligned} \quad (109)$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+) + f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u_{12}^* = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$\begin{aligned} u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-) &\stackrel{1.}{=} (u_{11}^* u_{21} - u_{22} u_{12}^*) f(\varepsilon_+) + u_{22} u_{12}^* \\ &\stackrel{2.,3.,4.}{=} 2u_{11} u_{21} f(\varepsilon_+) - u_{11} u_{21} \\ &= u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \quad (110)$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \rightarrow ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$\begin{aligned} \Rightarrow \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &= \Omega \int_{1.\text{BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \\ &= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \quad (111)$$

$$u_{11} u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \quad (112)$$

Now for the other expectation values:

$$\Rightarrow \langle b_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21}^* (2f(\varepsilon_+) - 1) \quad (113)$$

$$\begin{aligned} \langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle &= \dots = \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-)] \\ &\stackrel{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [|u_{12}|^2 - |u_{11}|^2] f(\varepsilon_-) + |u_{11}|^2 \\ &\stackrel{5.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [(1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2] \\ &\stackrel{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \end{aligned} \quad (114)$$

$$\Rightarrow \langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (115)$$

$$\Rightarrow \langle b_{i\uparrow}^\dagger b_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] [(1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2] \quad (116)$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \cos(k_y) + \mu) \quad (117)$$

$$\langle N^a \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2 \quad (118)$$

$$\langle N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2 \quad (119)$$

The constraint $\langle N_i^a \rangle = 1 = \langle N_i^b \rangle$ **should**, as gloal halffilling is already ensured and $\langle N^a \rangle$ and $\langle N^b \rangle$ are uniform by construction, be equivalent to $\langle N^a \rangle = \langle N^b \rangle$. Using the expressions above:

$$\langle N^a \rangle - \langle N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2)(1 - 2f(\varepsilon_+)) \stackrel{!}{=} 0 \quad (120)$$

Numerics say that $\langle N^a \rangle = \langle N^b \rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). **I feel like there is a more sophisticated argument to be made here, maybe even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1 - 2f(\varepsilon_+)$.**

Sanity check:

$$\langle N^a + N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} 1 = 2 \quad (121)$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (122)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2] \quad (123)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \quad (124)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_x/2} u_{11} u_{21}^* (2f(\varepsilon_+) - 1) \quad (125)$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. **I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.**

Another observation:

$$\begin{aligned} \frac{1}{2} \sum_{\sigma} \left(\langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle + \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle \right) &= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} \\ &= \frac{1}{2\pi} \int_{[-\pi, \pi]} dk_y e^{ik_y} \\ &= \frac{\sin(\pi)}{\pi} \\ &= 0 \end{aligned} \quad (126)$$

$$\Rightarrow \frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = -\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle, \quad (127)$$

as expected. The numerical results are

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = 0.19754 \quad (128)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = -0.19754 \quad (129)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = 0.19754 \quad (130)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = 0.19754 \quad (131)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_2\sigma} \rangle = -0.19754 \quad (132)$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \quad (133)$$

$$(134)$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}
\frac{\langle H \rangle}{N} &= \frac{1}{N} \left\langle \sum_{q\sigma} \varepsilon_{+\sigma} \alpha_{+\sigma}^\dagger \alpha_{+\sigma} + \varepsilon_{-\sigma} \alpha_{-\sigma}^\dagger \alpha_{-\sigma} \right\rangle \\
&\xrightarrow{\text{th. lim.}} \frac{1}{N} \frac{\Omega N}{2} \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} \varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\
&= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+ (2f(\varepsilon_+) - 1)
\end{aligned} \tag{135}$$

(the blue 2 in the second line is the spin degeneracy)

Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+ (2f(\varepsilon_+) - 1) \xrightarrow{\text{ph-trafo}} \varepsilon_- (2f(\varepsilon_-) - 1) = -\varepsilon_+ (1 - 2f(\varepsilon_+)) = \varepsilon_+ (2f(\varepsilon_+) - 1) \tag{136}$$

The full energy density is:

$$\frac{\langle H \rangle}{N} = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+(\vec{k}) (2f(\varepsilon_+(\vec{k})) - 1) + 6 \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) = -1.2206 J \tag{137}$$

4.3.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa, \chi, T=0)}{N} = \frac{E_0}{N} + \frac{1}{N} \sum_{\vec{k}} \varepsilon(\vec{k}) \tag{138}$$

$$E_0 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 \right) \tag{139}$$

$$\varepsilon(\vec{k}) = -2 \left| -\tilde{J} \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \right| \sqrt{\cos^2(k_1) + \cos^2(k_3) + \sin^2(k_2)} \tag{140}$$

From now on I use the shorthand $|\langle \chi_{ij} \rangle| \equiv \chi$.

$$\begin{aligned}
\Rightarrow f_0 &= \tilde{J} (6(1 - 6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma |\chi(1 - 4\tilde{\kappa}\chi^2)|) \\
&= \tilde{J} (\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4)
\end{aligned} \tag{141}$$

with

$$\gamma = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \sqrt{\cos^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.18524 \tag{142}$$

Compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates $f/M (f/2)!$

$$\frac{\partial f_0}{\partial \chi} = \tilde{J} (\pm 2\gamma - 12\chi) (12\tilde{\kappa}\chi^2 - 1) \quad \frac{\partial^2 f_0}{\partial \chi^2} = \tilde{J} (12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2) \tag{143}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \quad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{144}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6} \left(\frac{\tilde{\kappa}\gamma^2}{18} - 1 \right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J} \left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9} \sqrt{\frac{3}{\tilde{\kappa}}} \right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \quad (145)$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.6$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

$$\chi = \gamma/6 = 0.19754, \quad (146)$$

with the free energy

$$f_0 = -1.22206 J. \quad (147)$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

4.3.3 fermi surface

The dispersion relation reads

$$\varepsilon(\vec{q}) = 2|t| \sqrt{\cos^2(q_x) + \sin^2 \left(\frac{1}{2} (q_x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right)} \quad (148)$$

I will start by calculating the fermi surface $\varepsilon(\vec{q}) = 0$, which means that the expression under the square root must vanish. Using

$$\sin^2 \left(\frac{1}{2} (q_x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right) = 1 + \sin(q_x) \sin(\sqrt{3}q_y) \quad (149)$$

I obtain the following equation for the fermi surface:

$$\underbrace{1 + \cos^2(q_0^x)}_{\in[1,2]} = - \underbrace{\sin(q_0^x) \sin(\sqrt{3}q_0^y)}_{\in[-1,1]}. \quad (150)$$

The equation can be simplified by comparing the ranges of both sides (see above)

$$\Rightarrow 1 + \cos^2(q_0^x) \stackrel{!}{=} 1 \Rightarrow q_0^x = \frac{\pi}{2} + \pi k, \quad k \in \mathbb{Z} \quad (151)$$

$$\Rightarrow \sin(\sqrt{3}q_0^y) = \begin{cases} -1 & k \text{ even} \\ 1 & k \text{ odd} \end{cases} \quad (152)$$

The solution (and thereby the Fermi surface) is

$$q_0^x = \frac{\pi}{2} + \pi k \quad (153)$$

$$q_0^y = \frac{1}{\sqrt{3}} \left(\frac{(-1)^{k+1}\pi}{2} + 2\pi m \right), \quad (154)$$

where $k, m \in \mathbb{Z}$. This means that the fermi surface is a collection of isolated points. More scientifically spoken: the state is a semi-metal. Two of those Fermi-points lie in the first Brillouin zone:

$$\vec{K} = \frac{\pi}{2} \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (k = m = 0) \quad (155)$$

$$\vec{K}' = \frac{\pi}{2} \begin{pmatrix} -1 \\ 1/\sqrt{3} \end{pmatrix} \quad (k = -1, m = 0). \quad (156)$$

Mind you that these points are not equivalent! Evidently they are time-reversal partners.

4.3.4 small momentum expansion

To access the low-energy physics we must approximate the Hamiltonian around the fermi-points using

$$f(\vec{q}_0 + \delta\vec{q}) \approx f(\vec{q}_0) + (\nabla_{\vec{q}} f(\vec{q})|_{\vec{q}=\vec{q}_0}) \cdot \delta\vec{q}. \quad (157)$$

Around $\vec{K}$:

$$\cos(\vec{\delta}_1(\vec{K} + \vec{p})) \approx \cos(\vec{\delta}_1\vec{K}) - \sin(\vec{\delta}_1\vec{K})\vec{\delta}_1\vec{p} = -\vec{\delta}_1\vec{p} \quad (158)$$

$$\sin(\vec{\delta}_2(\vec{K} + \vec{p})) \approx \sin(\vec{\delta}_2\vec{K}) + \cos(\vec{\delta}_2\vec{K})\vec{\delta}_2\vec{p} = \vec{\delta}_2\vec{p} \quad (159)$$

$$\cos(\vec{\delta}_3(\vec{K} + \vec{p})) \approx \cos(\vec{\delta}_3\vec{K}) - \sin(\vec{\delta}_3\vec{K})\vec{\delta}_3\vec{p} = -\vec{\delta}_3\vec{p} \quad (160)$$

Use

$$\vec{\delta}_1\vec{K} = \frac{\pi}{2} \quad (161)$$

$$\vec{\delta}_2\vec{K} = 0 \quad (162)$$

$$\vec{\delta}_3\vec{K} = \frac{\pi}{2} \quad (163)$$

to obtain

$$\Rightarrow H_{\vec{K}} \approx 2t \sum_{p,\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} -\vec{\delta}_1\vec{p} & (-\vec{\delta}_3 - i\vec{\delta}_2)\vec{p} \\ (-\vec{\delta}_3 + i\vec{\delta}_2)\vec{p} & \vec{\delta}_1\vec{p} \end{pmatrix} \begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \quad (164)$$

And around $\vec{K}' = -\vec{K}$:

$$\Rightarrow H_{\vec{K}'} \approx 2t \sum_{p,\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} \vec{\delta}_1\vec{p} & (\vec{\delta}_3 - i\vec{\delta}_2)\vec{p} \\ (\vec{\delta}_3 + i\vec{\delta}_2)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix} \begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \quad (165)$$

This gives the dispersion relation (for both cones)

$$\begin{aligned} \varepsilon &\approx 2t \sqrt{p_x^2 + \frac{1}{4} (p_x + \sqrt{3}p_y)^2 + \frac{1}{4} (p_x - \sqrt{3}p_y)^2} \\ &= \sqrt{6}t \sqrt{p_x^2 + p_y^2} \\ &= \sqrt{6}t |\vec{p}| \end{aligned} \quad (166)$$

Thus, the fermi velocity is $v_F = \sqrt{6}t$. $\vec{K}$ and $\vec{K}'$ are time reversal partners, thus the two hamiltonians should map onto each other under time reversal. I would guess that $T = -C$ in k-space, where C is complex conjugation, which would be consistent with the form of the Hamiltonian.

4.3.5 Continuum Model/ Gradient Expansion

Next I will derive the low-energy Hamiltonian in real space, as a continuum theory (/field theory). The real space Hamiltonian reads (copied from above with the chemical potentials set to zero)

$$H = \frac{t}{2} \sum_{i\sigma} \left(a_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + a_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \\ \left. - b_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - b_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_3;\sigma} + b_{i\sigma}^\dagger a_{i-\delta_3;\sigma} \right) \quad (167)$$

The idea is to decompose the the real-space wavefunction by its momentum - one part is the momentum is the momentum of the dirac point, the other point describes a small deviation from the dirac point (small wavevector - i large wavelength - i slowly varying in real space). This justifies a gradient expansion of the "deviation field". **think of a better explanation for this. But I guess its correct in its spirit.**

$$a_{i\sigma} \sim e^{i\vec{K}\vec{R}_i} A_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i} A_{K'\sigma}(\vec{R}_i) \\ b_i \sim e^{i\vec{K}\vec{R}_i} B_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i} B_{K'\sigma}(\vec{R}_i) \quad (168)$$

$A_{K\sigma}, A_{K'\sigma}, B_{K\sigma}$ and $B_{K'\sigma}$ are slowly varying fermionic fields. I begin with the expansion around the K -point. **If the above expansion is correct, the cross terms should vanish. Why?** The prototypical method is

$$a_{i\sigma}^\dagger a_{i+\delta;\sigma} \sim e^{i\vec{K}\vec{\delta}} A_{K\sigma}^\dagger(\vec{R}_i) A_{K\sigma}(\vec{R}_i + \vec{\delta}) \\ \approx e^{i\vec{K}\vec{\delta}} A_{K\sigma}^\dagger(\vec{R}_i) \left(1 + \vec{\delta} \vec{\nabla}_{\vec{R}} \right) A_{K\sigma}(\vec{R}_i) \\ = e^{i\vec{K}\vec{\delta}} A_{K\sigma}^\dagger(\vec{R}_i) \left(1 + i\vec{\delta} \vec{p} \right) A_{K\sigma}(\vec{R}_i), \quad (169)$$

where $\vec{p} = -i\vec{\nabla}_{\vec{R}}$. The diagonal terms are

$$A_{K\sigma}^\dagger(\vec{R}_i) \left[e^{i\vec{K}\vec{\delta}_1} (1 + i\vec{\delta}_1 \vec{p}) + e^{-i\vec{K}\vec{\delta}_1} (1 - i\vec{\delta}_1 \vec{p}) \right] A_{K\sigma}(\vec{R}_i) \quad (170)$$

$$B_{K\sigma}^\dagger(\vec{R}_i) \left[-e^{i\vec{K}\vec{\delta}_1} (1 + i\vec{\delta}_1 \vec{p}) - e^{-i\vec{K}\vec{\delta}_1} (1 - i\vec{\delta}_1 \vec{p}) \right] B_{K\sigma}(\vec{R}_i). \quad (171)$$

The off-diagonal terms are

$$A_{K\sigma}^\dagger(\vec{R}_i) \frac{t}{2} \left[-e^{i\vec{\delta}_2 \vec{K}} (1 + i\vec{\delta}_2 \vec{p}) + e^{-i\vec{\delta}_2 \vec{K}} (1 - i\vec{\delta}_2 \vec{p}) + e^{i\vec{\delta}_3 \vec{K}} (1 + i\vec{\delta}_3 \vec{p}) + e^{-i\vec{\delta}_3 \vec{K}} (1 - i\vec{\delta}_3 \vec{p}) \right] B_{K\sigma}(\vec{R}_i) \quad (172)$$

$$B_{K\sigma}^\dagger(\vec{R}_i) \frac{t}{2} \left[e^{i\vec{\delta}_2 \vec{K}} (1 + i\vec{\delta}_2 \vec{p}) - e^{-i\vec{\delta}_2 \vec{K}} (1 - i\vec{\delta}_2 \vec{p}) + e^{i\vec{\delta}_3 \vec{K}} (1 + i\vec{\delta}_3 \vec{p}) + e^{-i\vec{\delta}_3 \vec{K}} (1 - i\vec{\delta}_3 \vec{p}) \right] A_{K\sigma}(\vec{R}_i) \quad (173)$$

Use

$$\vec{\delta}_1 \vec{K} = \frac{\pi}{2} \quad (174)$$

$$\vec{\delta}_2 \vec{K} = 0 \quad (175)$$

$$\vec{\delta}_3 \vec{K} = \frac{\pi}{2} \quad (176)$$

to obtain

$$H_{\vec{K}} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} -\vec{\delta}_1 \vec{p} & (-\vec{\delta}_3 - i\vec{\delta}_2) \vec{p} \\ (-\vec{\delta}_3 + i\vec{\delta}_2) \vec{p} & \vec{\delta}_1 \vec{p} \end{pmatrix} \Psi_{K\sigma}(\vec{R}), \quad (177)$$

with

$$\Psi_{K\sigma}(\vec{R}) = \begin{pmatrix} A_{K\sigma}(\vec{R}) \\ B_{K\sigma}(\vec{R}) \end{pmatrix}. \quad (178)$$

Maybe find a few things to write about Isospin? However this is probably not the real Isospin of theory yet as H is not diagonal.

Doing the same around $\vec{K}'$ gives

$$H_{\vec{K}'} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K'\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} \vec{\delta}_1 \vec{p} & (\vec{\delta}_3 - i\vec{\delta}_2) \vec{p} \\ (\vec{\delta}_3 + i\vec{\delta}_2) \vec{p} & -\vec{\delta}_1 \vec{p} \end{pmatrix} \Psi_{K'\sigma}(\vec{R}) \quad (179)$$

Maybe it is handy to write $\vec{\delta}_1 = \vec{\delta}_2 + \vec{\delta}_3$. When Comparing this to the results obtained by expanding in reciprocal space, keep in mind that here $\vec{p}$ is an operator. So technically, one would have to do a kind of fourier transform for the fermionic fields to find the exact representation as found in reciprocal space. This is why the Hamiltonian looks so similar, even though we are still in real space here.

Imagine we did this **small frequency?** Fourier expansion and we found the reciprocal space Hamiltonian (replace the momentum operator by its eigenvalue!), the corresponding disperion relation would then read

$$\varepsilon(\vec{p}) = \left| 2t \begin{pmatrix} \mp \vec{\delta}_3 \vec{p} \\ \vec{\delta}_2 \vec{p} \\ \mp \vec{\delta}_3 \vec{p} \end{pmatrix} \right| = \sqrt{6}t \sqrt{p_x^2 + p_y^2} = \sqrt{6}t |\vec{p}|, \quad (180)$$

which complies with the k-space result. In case I want to diagonalize this in operator form at some point; since the problem is non-interacting there must be a harmonic oscillator hidden in there.

4.4 0- π State on Strained Lattice

4.4.1 Basics of Strain

For the purpose of introducing strain into the model, I first establish its effect on the coordinates and exchange couplings **Awkward formulation**

$$\vec{R}_i \mapsto \vec{R}'_i = \vec{R}_i + \vec{U}(\vec{R}_i) \quad (181)$$

$$\vec{\delta}_{ij} \mapsto \vec{\delta}'_{ij} = \vec{\delta}_{ij} + \vec{U}(\vec{R}_j) - \vec{U}(\vec{R}_i) \quad (182)$$

$$\vec{\delta}_{ij} = \vec{R}_j - \vec{R}_i ;)$$

$\vec{U}(\vec{R})$ is the displacement field. We approximate that $\vec{U}$ varies slowly with $\vec{R}$:

$$\vec{U}(\vec{R}_j) = \vec{U}(\vec{R}_i + \vec{\delta}_{ij}) \approx \vec{U}(\vec{R}_i) + \left(\vec{\delta}_{ij} \vec{\nabla}_{\vec{R}} \right) \vec{U}(\vec{R})|_{\vec{R}_i} \quad (183)$$

$$(184)$$

It is convenient to redefine

$$\left(\vec{\delta}_{ij} \vec{\nabla}_{\vec{R}} \right) \vec{U}(\vec{R})|_{\vec{R}_i} = \underline{u} \cdot \vec{\delta}_{ij}, \quad (185)$$

where $u_{\alpha\beta} = \partial_\beta U_\alpha$ is the strain tensor.

$$\Rightarrow \vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u} \cdot \vec{\delta}_{ij} \quad (186)$$

I do the same approximation for the exchange couplings

$$J(\vec{\delta}_{ij}) \mapsto J(\vec{\delta}'_{ij}) \quad (187)$$

$$\approx J\left(\vec{\delta}_{ij} + \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (188)$$

$$\approx J(\vec{\delta}_{ij}) + \left(\underline{u} \cdot \vec{\delta}_{ij}\right) \cdot \left(\vec{\nabla}_{\vec{\delta}} J(\vec{\delta})|_{\vec{\delta}_{ij}}\right) \quad (189)$$

For the function $J(\vec{\delta})$ one usually uses the form

$$J(\vec{\delta}) = J \exp\left[-\beta\left(|\vec{\delta}|/a - 1\right)\right] \approx J\left(1 - \beta\left(|\vec{\delta}|/a - 1\right)\right), \quad (190)$$

where a is the lattice constant and $\beta = -d \ln(J)/d \ln(|\vec{\delta}|)_a$ the magneto-elastic coupling. This form is essentially motivated by the large U-limit of the Hubbard model. In this limit the exchange coupling is proportional to the overlap of the orbitals of neighboring atoms. In this sense one can approximate those as two exponential functions, that are decreasing away from the position of their respective atoms. Therefore, their overlap also decreases exponentially.

$$\begin{aligned} &\Rightarrow \vec{\nabla}_{\vec{\delta}} J(\vec{\delta})|_{\vec{\delta}_{ij}} \stackrel{|\vec{\delta}_{ij}|=a}{=} -\frac{J\beta}{a^2} \vec{\delta}_{ij} \\ &\Rightarrow J(\vec{\delta}'_{ij}) \approx J(\vec{\delta}_{ij}) - \frac{J\beta}{a^2} \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \end{aligned} \quad (191)$$

Looking at the matrix elements (170 and below), one also requires the expansion of $\exp\left[i\vec{\delta}'_{ij}\vec{K}\right]$. To summarize:

$$\vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u} \cdot \vec{\delta}_{ij} \quad (192)$$

$$J(\vec{\delta}'_{ij}) \approx J - \frac{J\beta}{a^2} \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (193)$$

$$e^{i\vec{K} \cdot \vec{\delta}'_{ij}} \approx e^{i\vec{K} \cdot \vec{\delta}_{ij}} + ie^{i\vec{K} \cdot \vec{\delta}_{ij}} \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (194)$$

In this approximation the Mean field does not react to the strain at all??

4.4.2 Straining the 0- π -State

The matrix elements of the strained Hamiltonian are of the form

$$\frac{1}{2} \left[J(\vec{\delta}'_{ij}) e^{i\vec{K} \cdot \vec{\delta}'_{ij}} (1 + i\vec{\delta}'_{ij} \cdot \vec{p}) + J(-\vec{\delta}'_{ij}) e^{-i\vec{K} \cdot \vec{\delta}'_{ij}} (1 - i\vec{\delta}'_{ij} \cdot \vec{p}) \right] \quad (195)$$

I expand in strain but keep only terms of zeroth order in strain and field gradient as well as terms of either first order in strain or first order in field-gradient:

$$J(\vec{\delta}'_{ij}) e^{i\vec{K} \cdot \vec{\delta}'_{ij}} (1 + i\vec{\delta}'_{ij} \cdot \vec{p}) \approx J e^{i\vec{K} \cdot \vec{\delta}_{ij}} (1 + i\vec{\delta}_{ij} \cdot \vec{p}) + iJ e^{i\vec{K} \cdot \vec{\delta}_{ij}} \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) - \frac{J\beta}{a^2} e^{i\vec{K} \cdot \vec{\delta}_{ij}} \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (196)$$

How do different orders of strain and field gradients compare??

The first term is the zeroth order. It turns out that when adding up two terms corresponding to opposite directions the first term in first order acquires a factor of 2 while the second term vanishes. **The cancelling of the second term is a Bravais lattice effect that is not present in e.g. Graphene! The second term in first order would have been imaginary, what is the significance of that?**

Using

$$\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} = \vec{\delta}_{ij} \cdot \underline{u}^T \cdot \vec{K}, \quad (197)$$

I rename $\vec{R}' \rightarrow \vec{R}$ and get

$$H_{\vec{K}} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} -\vec{\delta}_1 \left(\vec{p} + \underline{u}^T \cdot \vec{K} \right) & \left(-\vec{\delta}_3 - i\vec{\delta}_2 \right) \left(\vec{p} + \underline{u}^T \cdot \vec{K} \right) \\ \left(-\vec{\delta}_3 + i\vec{\delta}_2 \right) \left(\vec{p} + \underline{u}^T \cdot \vec{K} \right) & \vec{\delta}_1 \left(\vec{p} + \underline{u}^T \cdot \vec{K} \right) \end{pmatrix} \Psi_{K\sigma}(\vec{R}), \quad (198)$$

Doing the same procedure around $\vec{K}' = -\vec{K}$ I get

$$H_{\vec{K}'} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K'\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} \vec{\delta}_1 \left(\vec{p} + \underline{u}^T \cdot \vec{K}' \right) & \left(\vec{\delta}_3 - i\vec{\delta}_2 \right) \left(\vec{p} + \underline{u}^T \cdot \vec{K}' \right) \\ \left(\vec{\delta}_3 + i\vec{\delta}_2 \right) \left(\vec{p} + \underline{u}^T \cdot \vec{K}' \right) & -\vec{\delta}_1 \left(\vec{p} + \underline{u}^T \cdot \vec{K}' \right) \end{pmatrix} \Psi_{K'\sigma}(\vec{R}) \quad (199)$$

therefore the effective vector potential is

$$\vec{A}_{\vec{K}} = -\vec{A}_{\vec{K}'} \quad (200)$$

$$= \underline{u}^T \cdot \vec{K} \quad (201)$$

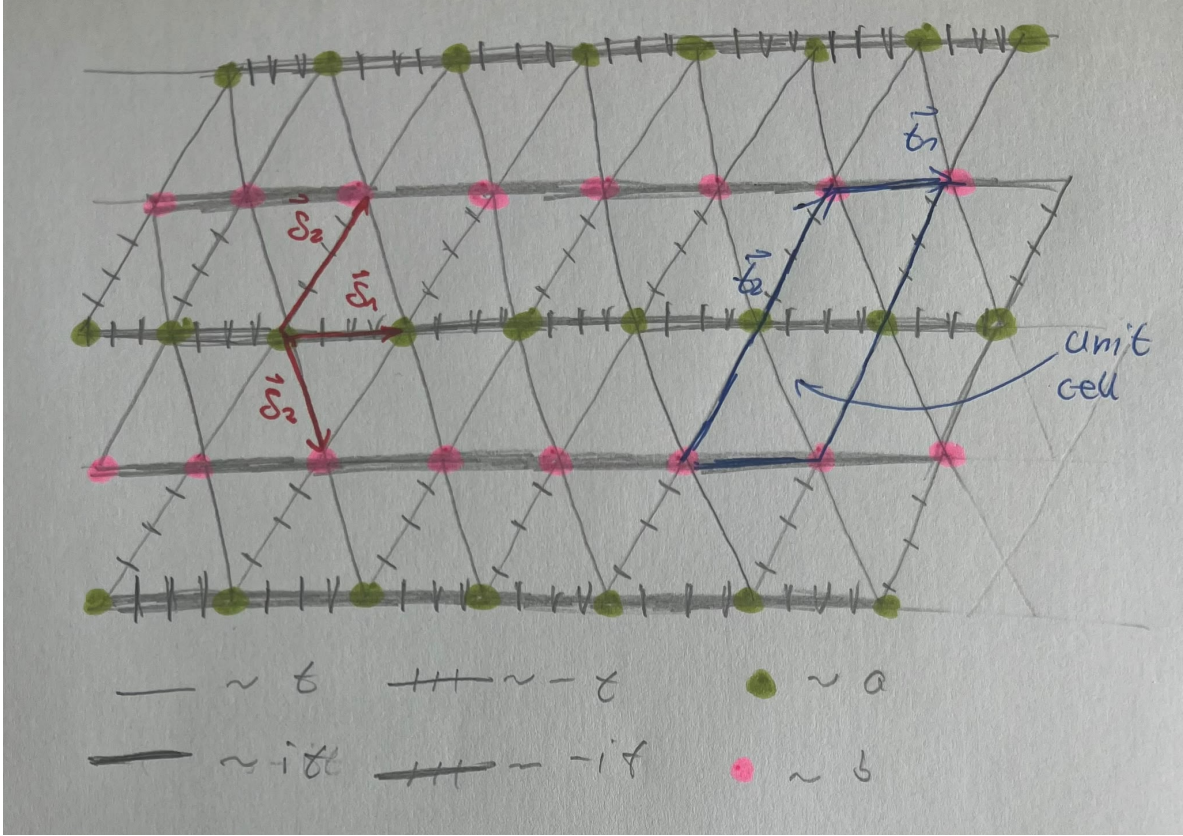
$$= \frac{\pi}{2} \begin{pmatrix} u_{xx} - \frac{u_{yx}}{\sqrt{3}} \\ u_{xy} - \frac{u_{yy}}{\sqrt{3}} \end{pmatrix}. \quad (202)$$

Sadly, this vector potential produces no pseudo-magnetic field. **What about higher orders?**

$$\vec{B} = (\partial_x A_y - \partial_y A_x) \hat{e}_z \quad (203)$$

Judging from the form of the Hamiltonian one would have a hard time introducing a $\vec{\delta}_{ij}$ -dependent shift of the momentum in any case. Maybe there is a smart argument why this can't be achieved for a Bravais lattice.

4.5 $\pi/2$ - $\pi/2$ and $-\pi/2$ - $-\pi/2$ States



4.5.1 Pedestrian Method

This is very similar to the $\pi - 0$ -flux-state. The bare Hamiltonian is

$$H = \sum_{i,\sigma} \left(\frac{t}{2} \left[-ia_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + ia_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \right. \\ \left. \left. + ib_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - ib_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right] \right. \\ \left. + \mu_a a_{i\sigma}^\dagger a_{i\sigma} + \mu_b b_{i\sigma}^\dagger b_{i\sigma} \right) - \frac{N(\mu_a + \mu_b)}{2} \quad (204)$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices.

$$\Rightarrow H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t \sin(\vec{q}\vec{\delta}_1) + \mu_a & 2t \cos(\vec{q}\vec{\delta}_3) - 2it \sin(\vec{q}\vec{\delta}_2) \\ 2t \cos(\vec{q}\vec{\delta}_3) + 2it \sin(\vec{q}\vec{\delta}_2) & -2t \sin(\vec{q}\vec{\delta}_1) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \quad (205)$$

Maybe I should show explicitly that $\mu = 0$, I guess $\mu_a = -\mu_b = \mu$ is clear

Using $\mu_a = \mu_b = 0$:

$$\vec{h} = 2t \begin{pmatrix} \cos(\vec{q}\vec{\delta}_1) \\ \sin(\vec{q}\vec{\delta}_2) \\ \sin(\vec{q}\vec{\delta}_3) \end{pmatrix} \mapsto 2t \begin{pmatrix} \cos(k_y - k_x/2) \\ \sin(k_x/2) \\ \sin(k_y) \end{pmatrix} \quad (206)$$

$$\varepsilon(\vec{k})_{\pm} = \pm 2|t| \sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \quad (207)$$

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t \sin(k_y))}} \begin{pmatrix} \varepsilon_+ + 2t \sin(k_y) & -2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) \\ 2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) & \varepsilon_+ + 2t \sin(k_y) \end{pmatrix} \quad (208)$$

$$u_{11}u_{21} = (t/\varepsilon_+)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \quad (209)$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \sin(k_y)) \quad (210)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (211)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2] \quad (212)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11}u_{21}(2f(\varepsilon_+) - 1) \quad (213)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+) - 1) \quad (214)$$

One has to be a bit careful of where the sign of t enters when choosing $t = 1$ for the self consistency. Due to the complex conjugation the σ_z -terms are insensitive to the sign of t , but the (real) σ_x - and σ_y -terms are not. The results (choosing $t = 1$ are):

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = -0.200169i \quad (215)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i-\delta_1\sigma} \rangle = 0.200169i \quad (216)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = 0.200169i \quad (217)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i-\delta_1\sigma} \rangle = -0.200169i \quad (218)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = 0.200169 \quad (219)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_3\sigma} \rangle = 0.200169 \quad (220)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = 0.200169 \quad (221)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_2\sigma} \rangle = -0.200169 \quad (222)$$

This complies with the cartoon on top. The value for t is

$$|t_{ij}| = \left| J \langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2) \right| \stackrel{\kappa=10}{=} 0.88020 J. \quad (223)$$

The free energy is

$$\frac{\langle H \rangle}{N} = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+(\vec{k}) (2f(\varepsilon_+(\vec{k})) - 1) + 6 \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) = -1.24978 J \quad (224)$$

4.5.2 Pedro's Method

The expression for the free energy is identical to the $0 - \pi$ -state except that γ gets replaced by a different constant α :

$$\alpha = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.20102 \quad (225)$$

Crucially α is larger than gamma. The results for this method are

$$|\chi| = \alpha/6 = 0.20017, \quad (226)$$

$$f_0 = \tilde{J} \left(\frac{1}{4\tilde{\kappa}} - \frac{2\alpha}{9} \sqrt{\frac{3}{\tilde{\kappa}}} \right) = -1.24979 J \quad (227)$$

The methods agree.

4.5.3 Chern number

Here I will calculate the Chern number associated with the state. The relevant formula for a 2-band system is

$$C = \frac{1}{4\pi} \int_{1.BZ} d^2 q \hat{h} \cdot \left(\partial_{q_x} \hat{h} \times \partial_{q_y} \hat{h} \right), \quad (228)$$

where $\hat{h}$ is the normalized Bloch vector ([is this the correct terminology?](#)).

$$\vec{h} = 2t \begin{pmatrix} \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \\ \sin\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) \\ \sin(q_x) \end{pmatrix} \equiv 2t \begin{pmatrix} \cos(\delta_3) \\ \sin(\delta_2) \\ \sin(\delta_1) \end{pmatrix}, \quad |\vec{h}| \equiv \varepsilon(\vec{q}) \quad (229)$$

I first simplify [228](#) a bit using

$$\partial_\alpha \hat{h} = \frac{1}{|\vec{h}|} \left(\partial_\alpha \vec{h} - (\hat{h} \cdot \partial_\alpha \vec{h}) \hat{h} \right). \quad (230)$$

When applying this to [228](#), 3 out of the for terms that appear are proportional to $\hat{h} \times \hat{h} = \vec{0}$.

$$\Rightarrow C = \frac{1}{4\pi} \int_{1.BZ} d^2 q [\varepsilon(\vec{q})]^{-3} \vec{h} \cdot \left(\partial_{q_x} \vec{h} \times \partial_{q_y} \vec{h} \right) \quad (231)$$

$$\partial_{q_x} \vec{h} = t \begin{pmatrix} -\sin(\delta_3) \\ \cos(\delta_2) \\ 2\cos(\delta_1) \end{pmatrix}, \quad \partial_{q_y} \vec{h} = \sqrt{3}t \begin{pmatrix} \sin(\delta_1) \\ \cos(\delta_2) \\ 0 \end{pmatrix} \quad (232)$$

$$\begin{aligned}
\Rightarrow \vec{h} \cdot \left(\partial_{q_x} \vec{h} \times \partial_{q_y} \vec{h} \right) &= 4\sqrt{3}t^3 \left(-\cos(\delta_1) \cos(\delta_2) \cos(\delta_3) + \cos(\delta_1) \sin(\delta_2) \sin(\delta_3) - \sin(\delta_1) \cos(\delta_2) \sin(\delta_3) \right) \\
&= 4\sqrt{3}t^3 \left(\sin(\delta_1) \sin(\delta_2) \cos(\delta_3) - 1 \right)
\end{aligned} \tag{233}$$

$$\Rightarrow C = \frac{\sqrt{3}}{8\pi} \int_{\text{1.BZ}} d^2q \frac{\sin(\delta_1) \sin(\delta_2) \cos(\delta_3) - 1}{\left(\sin^2(\delta_1) + \sin^2(\delta_2) + \cos^2(\delta_3) \right)^{3/2}} \tag{234}$$

For evaluation I use the transformation:

$$\begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{235}$$

$$\Rightarrow C = \frac{1}{8\pi} \int_{[-\pi, \pi]^2} d^2k \frac{\sin(k_y) \sin(k_y - k_x/2) \cos(k_x/2) - 1}{\left(\sin^2(k_y) + \sin^2(k_y - k_x/2) + \cos^2(k_x/2) \right)^{3/2}} \tag{236}$$

The result is $C = -1$.

References

- [1] Mary Madelynn Nayga, Stephan Rachel, and Matthias Vojta. “Magnon Landau Levels and Emergent Supersymmetry in Strained Antiferromagnets”. In: *Phys. Rev. Lett.* 123 (20 Nov. 2019), p. 207204. DOI: [10.1103/PhysRevLett.123.207204](https://doi.org/10.1103/PhysRevLett.123.207204). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.123.207204>.

Part III

other

5. mail 10.07

$$\mathcal{H} = \frac{J_H}{M'} \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j + \frac{\kappa}{(M')^2} \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \quad (237)$$

transformations:

$$\text{Eric : } \vec{S}_i = \sum_{\alpha\beta} f_{i\alpha}^\dagger \frac{\vec{\sigma}}{2} f_{i\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{1}{4} - \frac{1}{2} \sum_{\alpha\beta} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} \quad (238)$$

$$\text{Pedro : } S_i^{\beta\alpha} = f_{i\beta}^\dagger f_{i\alpha} - \frac{Q}{M} \delta_{\alpha\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{M}{4} - \sum_{\alpha\beta} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} \quad (239)$$

difference of factor 2 for $M = 2$, which is my case. (Also, for me $M' = 1$ while Pedro used $M' = M$.)
mean field Hamiltonians:

$$\text{Eric : } \mathcal{H} \approx -J \sum_{\langle ij \rangle} \left[(1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c.} \right] \quad (240)$$

$$+ 2(-1 - \kappa/2 + 6\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \quad (241)$$

$$\text{Pedro : } \mathcal{H} \approx -J \sum_{\langle ij \rangle} \left[2 + \boxed{2\kappa} - 16\kappa |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger (f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow}) \right) + \text{h.c.} \right] \quad (242)$$

$$+ 2(-2 \boxed{-2\kappa} + 24\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \quad (243)$$

Every term without a κ (coming from the bilinear exchange term) is off by a factor of 2. Every term with a κ (coming from the biquadratic exchange term) is off by a factor of 4. Hence, the only problem is the factor 2 in the fermionic representation. The boxed terms don't appear in Pedro's notes as they go like M^{-1} and are therefore irrelevant in the large M limit.